

AM Lab

EE 451: Communications Systems — University of Scranton

Based on the lab, "Baba Yaga's Hut: A Laboratory Experience in Phasor Analysis," written by Kristina Collins KD8OXT and David Kazdan AD8Y for the Case Western Reserve University EECS 351 class. That lab was originally inspired by the work of Alan Wolke, W2AEW. His phasor demonstration, on which ours is based, can be viewed at <https://www.youtube.com/watch?v=5GGD99Qz1PA>. Adapted for the University of Scranton by Nathaniel Frissell W2NAF with assistance from Claude.AI.

Abstract

This lab will help you get familiar with amplitude modulated signals. We will use four tools to observe signals simultaneously: an oscilloscope (time domain), a spectrum analyzer (frequency domain), an XY phasor display (built from a homodyne receiver using MiniCircuits¹ components), and an AM radio (demodulated audio).

1 Introduction

In this lab we examine an AM signal simultaneously in three domains — time, frequency, and phasor — while also listening to the demodulated audio on an AM radio. The classic AM envelope is shown in Figure 1: a high-frequency carrier whose amplitude varies slowly with the modulating signal, producing the characteristic “fuzzy football” shape.



Figure 1: The USPS Amateur Radio stamp showcases the classic face of an AM radio signal in the time domain: a stationary envelope with the modulation fuzz inside. This is what you should see on Oscilloscope 1.

The transmitter is a Siglent SDG2122X waveform generator configured for AM modulation at a carrier frequency of 1260 kHz. A coaxial cable carries the modulated signal to the receiver, simulating a transmitter, free-space path, and receiver antenna.

¹cf. <https://www.minicircuits.com/>.

The receiver is a direct-conversion (homodyne) system built from a second waveform generator and MiniCircuits components (a splitter, two mixers, and two low-pass filters). We call this “Baba Yaga’s hut” — in Russian folklore, Baba Yaga² was a witch who lived in a hut that stood on fowl’s legs; at her command, the hut would spin and fly. Our laboratory version includes a “glass floor” (the oscilloscope XY display) through which you observe the spinning phasor, and a frequency knob that lets you lock onto the carrier and watch the modulation unfold beneath your feet (Figure 2).

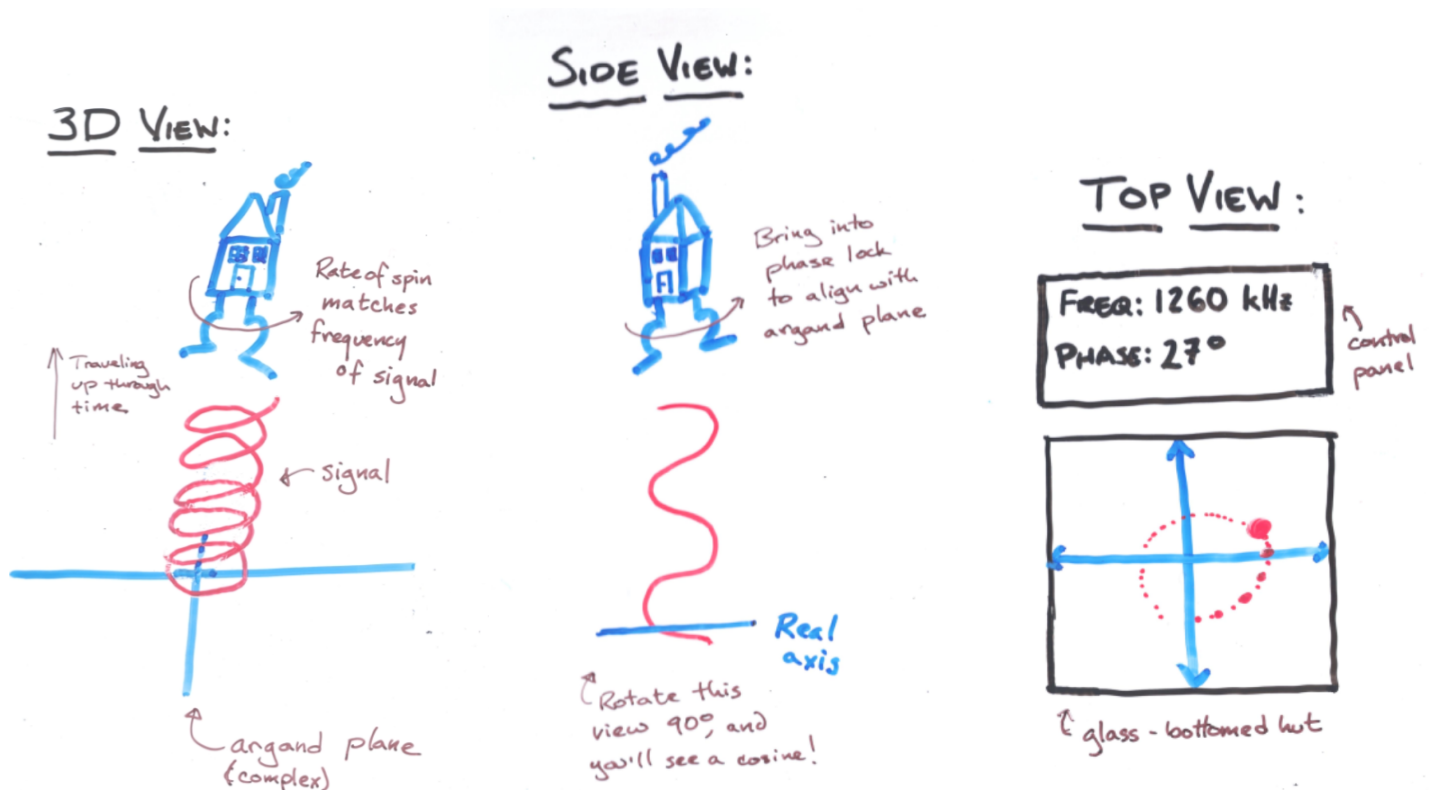


Figure 2: Illustration of the hut metaphor. Left: 3D view — the analytic signal’s rotating vector is on the lab floor, and you fly over it in the hut. Center: side view — bring the hut into phase lock with the carrier. Right: top view (through the glass floor) — what remains is the phasor, from which modulation can be observed.

The receiver splits the incoming signal and mixes it with a local oscillator at the carrier frequency. One mixer uses the LO directly (in-phase, I); the other uses a 90°-shifted copy (quadrature, Q). After low-pass filtering, the I and Q baseband outputs are fed to the X and Y inputs of an oscilloscope in XY mode, producing a real-time phasor display. The schematic is shown in Figure 3.

²For an academic analysis of her folklore, see ISBN 978-0820467696.

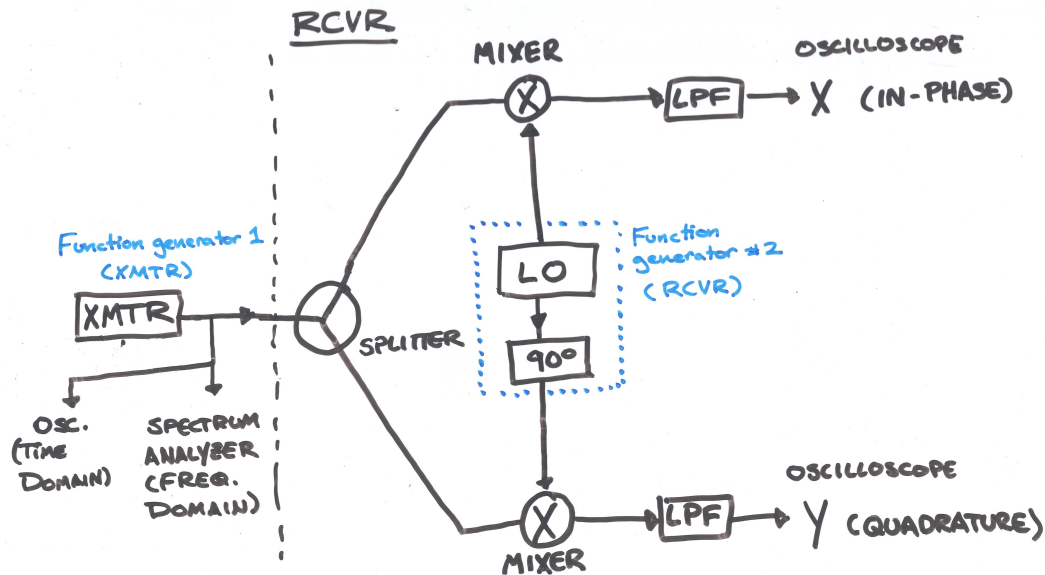


Figure 3: Schematic for the apparatus. The receiving function generator and the MiniCircuits components form a direct-conversion (homodyne) receiver.

The following section walks you through configuring each instrument. Work through it in order before beginning the experiments.

2 Equipment Setup

The lab bench has been **wired** for you, but you must **configure every instrument** before you begin. Work through each subsection below in order. Do not skip ahead — later instruments depend on earlier ones being configured correctly.

2.1 Verify Cabling

Before touching any settings, confirm every cable is connected. Use Figure 3 and Table 1 as your reference.

From	Through	To	Notes
XMTR "10 MHz Out"	BNC cable	RCVR "10 MHz In"	Critical. Shares the reference clock. Without it, the phasor spins uncontrollably.
XMTR CH1	ZSC-2-2+ power splitter	Spectrum Analyzer + Oscilloscope 1	50 Ω matching and port isolation.
XMTR CH2	BNC Tee	Mixer 1 (I) + Mixer 2 (Q)	Passive split to both mixers in the direct-conversion receiver.
RCVR CH1	BNC cable	Mixer 1 (LO)	In-phase local oscillator (0°).
RCVR CH2	BNC cable	Mixer 2 (LO)	Quadrature local oscillator (90°).
Mixer 1 IF out	Low-pass filter	Osc. 2 CH1 (X)	In-phase baseband $\rightarrow$ phasor X-axis.
Mixer 2 IF out	Low-pass filter	Osc. 2 CH2 (Y)	Quadrature baseband $\rightarrow$ phasor Y-axis.
Osc. 1	BNC Tee	Wire spool near AM radio	RF coupling to AM radio antenna for demodulated audio.

Table 1: Signal routing (should already be wired).

2.2 Function Generator #1 — Transmitter (XMTR)

This Siglent SDG2122X serves as the transmitter and the reference clock for the entire lab.

Setting	Notes
Utility ⇨ Clock Source: Internal	Func Gen #1 serves as the reference clock.
Utility ⇨ CH Copy Coupling Track: On Phase Dev: 0°	This forces both channel 1 and 2 to output signals identical in both frequency and phase.
Parameter Frequency: 1260 kHz Amplitude: 1 V_{pp} Offset: 0 Vdc Phase: 0°	1260 kHz places us in the AM broadcast band (560–1600 kHz). Carrier amplitude is varied to 0.5 V_{pp} during the carrier voltage experiment. Phase is varied 0°–90° during the phase experiment.
Mod Type: AM Source: Internal AM Depth: 75% AM Freq: 440 Hz Shape: Sine Turn “Output” buttons above both channel BNC ports on. They should both be lit up green.	These are the default starting parameters. During the lab, you will vary: • Depth: 0%, 50%, 100%, 120% (Table 8) • Frequency: 1 Hz, 440 Hz, 1 kHz, 5 kHz (Table 9) • Type: Switch from AM (DSB-FC) to DSB-AM (DSB-SC) for Section 8 (Table 10)

Table 2: XMTR settings (Func Gen #1).

2.3 Function Generator #2 — Receiver (RCVR)

This Siglent SDG2122X provides the local oscillator for the direct-conversion receiver.

Setting	Notes
Utility ⇨ Clock Source: External	Func Gen #2 gets its clock from Func Gen #1.
Utility ⇨ CH Copy Coupling Track: On PhaseDev: 90°	This forces both channel 1 and 2 to output identical in frequency but 90° offset in phase. Channel #1 becomes the In-Phase local oscillator, while Channel #2 becomes the quadrature local oscillator.
Parameter Frequency: 1260 kHz Amplitude: 1 V _{pp} Offset: 0 Vdc Phase: Vary for Calibration	Frequency must match FG1 exactly. Even a few Hz difference will cause the phasor to spin instead of staying locked. The shared 10 MHz reference clock handles frequency synchronization, but make sure the dial reads the same value (1260 kHz). Start with the phase at 0°, but you will vary this to “calibrate” the phasor view so that the I channel is strictly along the x-axis (see Section 4).
Turn “Output” buttons above both channel BNC ports on. They should both be lit up green.	

Table 3: RCVR settings (Func Gen #2).

2.4 Spectrum Analyzer — Frequency Domain View

Siglent SSA 3021X Plus.

Setting	Notes
Frequency button Start Freq: 1.25 MHz Stop Freq: 1.27 MHz	These give a 20 kHz span centered on the 1.260 MHz carrier. At the default 440 Hz modulating frequency, you should see a tall carrier spike at 1.260 MHz flanked by two smaller sideband spikes at 1.25956 MHz and 1.26044 MHz (carrier $\pm$ 440 Hz). When varying modulating frequency (Table 9): The sidebands move further out at higher modulating frequencies. At 5 kHz the sidebands are at $\pm$5 kHz — near the edges of this 20 kHz span. Widen the span (e.g., Start: 1.24 MHz, Stop: 1.28 MHz) for the 5 kHz experiment.
Amplitude button Ref Level: 0 dBm Scale/Div: 10 dB	Ref Level sets the top gridline of the display. 0 dBm with 10 dB/div across 10 vertical divisions gives 100 dB of dynamic range on screen (0 to –100 dBm). The carrier spike should appear in the top portion of the display. Adjust Ref Level up or down if the carrier is clipped or too small.
BW button RBW: 100 Hz VBW: 100 Hz	RBW (Resolution Bandwidth): The bandwidth of the bandpass filter used to sweep and measure signals. The narrower the RBW, the finer the resolution (and the longer the sweep time). At 100 Hz RBW, you can clearly resolve sidebands as close as 440 Hz from the carrier. VBW (Video Bandwidth): A post-detection low-pass filter that smooths the trace and reduces noise on the display. Lowering the VBW makes the trace cleaner but increases sweep time. Turn off Auto RBW and Auto VBW to set these manually.
PEAK button Peak Search	Places a marker on the strongest signal (the carrier). Use Next Peak to hop to the sidebands. The marker readout gives exact frequency and amplitude — useful for measuring sideband levels relative to the carrier and verifying the expected $\pm f_{\text{mod}}$ spacing.
Trace (optional) Type: Clear Write Detector: Positive Peak	Default settings. Clear Write redraws the trace each sweep. If the trace looks noisy, you can switch to Average mode (e.g., 10 averages) for a smoother display.

Table 4: Spectrum analyzer settings.

2.5 Oscilloscope 1 — Time Domain View

Tektronix TDS 2002 (10 horizontal $\times$ 8 vertical divisions).

Setting	Notes
Horizontal (Timebase) SEC/DIV: 500 μ s	500 μ s/div gives ~ 5 ms on screen (10 divs), showing ~ 2 full modulation envelope cycles at 440 Hz ($T = 2.27$ ms). Adjust for other modulating frequencies: • 1 Hz $\rightarrow$ 200 ms/div • 1 kHz $\rightarrow$ 200 μs/div • 5 kHz $\rightarrow$ 50 μs/div Goal: ~ 2 envelope cycles on screen.
Vertical VOLTS/DIV: 200 mV Coupling: AC BW Limit: Off Probe: 1X	IMPORTANT: Set Probe to 1X (CH1 Menu $\mapsto$ Probe $\mapsto$ 1X). We are feeding the scope with a BNC cable, not a 10X probe. If left on 10X, the displayed voltage will read $10\times$ too high. Expected signal level at default settings (1 V_{pp} carrier, 75% AM, 440 Hz): The ZSC-2-2+ power splitter has ~ 3 dB insertion loss, halving the power and reducing the carrier to $\sim 0.7 V_{pp}$. With 75% AM depth, the envelope peak is $(1 + m) = 1.75\times$ the carrier, giving $\sim 1.2 V_{pp}$ at the scope input — about 6 divisions at 200 mV/div. AC vs DC Coupling: Coupling refers to what the scope allows through to the display. DC coupling passes everything — the signal plus any constant DC offset. AC coupling inserts a high-pass filter (a series capacitor inside the scope) that blocks DC and very low frequencies, passing only the AC signal. We use AC coupling here because the splitter can introduce small DC offsets that would shift the waveform off-center on screen. One caveat: AC coupling will distort very low-frequency signals (below ~ 10 Hz), so at 1 Hz modulating frequency the envelope may appear slightly tilted. If you see that, switch to DC coupling temporarily. Increase VOLTS/DIV when modulation depth goes to 100–120%. Decrease when carrier amplitude is reduced (e.g., 0.5 V_{pp} experiment).
Trigger Type: Edge Source: CH1 Slope: Rising Mode: Normal Coupling: LF Reject Level: Just below positive envelope peak	What the trigger does: An oscilloscope continuously samples the input, but it needs to know <i>when</i> to start drawing each sweep across the screen. The trigger is that starting gun. When the signal crosses the trigger level in the specified direction (rising or falling), the scope begins a new sweep. If it triggers at the same point every time, successive sweeps overlap perfectly and you see a stable picture. If it triggers at different points, the display jitters or “rolls.” Trigger Coupling (not the same as channel coupling!) The channel coupling (CH1 Menu) affects the <i>displayed</i> waveform. Trigger coupling affects only what the <i>trigger circuit</i> sees — it does not change what’s shown on screen. We recommend LF Reject , which high-pass filters the trigger path, removing DC and low-frequency drift while passing the 1260 kHz carrier intact. Produces the cleanest triggering for this lab. Avoid HF Reject — it strips out the carrier and produces poor results. Why trigger level matters for AM: Our signal has two periodic components — the fast 1260 kHz carrier and the slow 440 Hz modulation envelope. We want to trigger once per <i>envelope</i> cycle, not once per carrier cycle. The trick: set the trigger level just below the envelope peak. Only the tallest carrier cycles (near the envelope maximum) will reach that level, so the scope triggers once per modulation period and the envelope appears stable. If the level is too low, multiple carrier cycles cross it per envelope period and the display smears. If the display jitters, nudge the trigger level up. You may need to re-adjust whenever the modulation depth or carrier amplitude changes.

Table 5: Oscilloscope 1 settings (time domain).

2.6 Oscilloscope 2 — Phasor Domain View

Tektronix TDS 2002 in XY mode. In this mode there is no time sweep and no trigger — the scope plots CH1 (In-Phase) vs. CH2 (Quadrature) continuously.

Setting	Notes
Display Format: XY	Critical. This switches the scope from the normal time-sweep (YT) mode to XY mode, where CH1 drives the horizontal axis and CH2 drives the vertical axis. On the TDS 2002, press the Display button and set Format to XY. What you're seeing: The direct-conversion receiver mixes the incoming signal with a local oscillator (I channel) and a 90°-shifted copy (Q channel), then low-pass filters each output. The result is the baseband I and Q components of the modulation. Plotting I vs Q gives you the phasor — a real-time view of the signal's amplitude and phase relative to the carrier.
CH1 (X-axis = In-Phase) VOLTS/DIV: 50 mV Coupling: DC Probe: 1X	Probe: 1X — same as Oscilloscope 1, we are using BNC cables not probes. DC coupling is essential here. Unlike Oscilloscope 1, we must use DC coupling on the phasor scope. The carrier component of the received signal appears as a DC offset in the baseband after mixing down. This DC offset is what positions the phasor dot away from the origin. If you use AC coupling, the scope's high-pass filter will remove this DC component and the phasor will be incorrectly centered at the origin. Expected signal levels: The mixer (ZAD-6+) has ~6 dB conversion loss, and the signal arrives via a BNC tee split. The baseband output will be in the tens to hundreds of mV range. Start at 50 mV/div and adjust so the phasor fills roughly half the screen.
CH2 (Y-axis = Quadrature) VOLTS/DIV: 50 mV Coupling: DC Probe: 1X	VOLTS/DIV must match CH1. Both axes must have the same scale, otherwise the phasor diagram will be stretched or squished in one direction — a circle would appear as an ellipse, and angles would be distorted. Always change both channels together. DC coupling and Probe 1X for the same reasons as CH1.
Persistence Display ↔ Persist: 5 s	Persistence keeps previous phasor positions on screen, building up a trace over time. 5 seconds is a good default — long enough to see the full phasor motion at most modulating frequencies, while old traces fade naturally so you don't need to manually clear between settings changes. Switch to Infinite if you need to see the complete trace at very low modulating frequencies (e.g., 1 Hz, where the phasor takes a full second to sweep end to end and 5 s may not fully capture the motion). Toggle persistence to Off and back to Infinite to clear old traces.

Table 6: Oscilloscope 2 settings (phasor domain).

2.7 AM Radio

There is an AM radio on the bench with a spool of wire nearby that acts as an antenna coupling from Oscilloscope 1 (see Table 1). Tune the radio to **1260 kHz** (or as close as the dial allows). Turn the radio on and adjust the volume to a comfortable level. You should hear the 440 Hz modulation tone. If not, verify that the wire spool is near the radio's antenna and that the transmitter outputs are on.

2.8 Setup Checklist

Confirm the following before proceeding to the experiments:

- ☐ All cables connected per Table 1.
- ☐ XMTR and RCVR settings match Tables 2–3; all four outputs lit green.
- ☐ Spectrum analyzer shows carrier at 1.260 MHz with ± 440 Hz sidebands.
- ☐ Oscilloscope 1 shows an AM envelope (the classic “stamp” shape, Figure 1).
- ☐ Oscilloscope 2 shows a dot or line (you will calibrate this in Section 4).
- ☐ AM radio tuned to 1260 kHz; 440 Hz tone audible.
- ☐ Name and date on this document.

3 Observing the AM Signal

Now that all instruments are configured, we will observe the AM signal in each of the three domains and verify it matches expectations.

3.1 Time Domain

Look at Oscilloscope 1. You should see the classic AM envelope shape shown in Figure 1: a 1260 kHz carrier oscillation whose amplitude varies at the 440 Hz modulation rate, producing a slowly expanding and contracting outline (the envelope) with the carrier “fuzz” inside. If the display is unstable, adjust the trigger level upward until you get a clean, repeating waveform.

Q3.1-1: Sketch or screenshot the waveform and note your oscilloscope settings (SEC/DIV, VOLTS/DIV, trigger level). (5 points.)

3.2 Frequency Domain

Look at the spectrum analyzer. You should see the carrier spike at 1.260 MHz flanked by two sideband spikes. Use the **Peak Search** button to place a marker on the carrier, then **Next Peak** to hop to the sidebands. Verify that the sidebands are at ± 440 Hz from the carrier.

Q3.2-1: Sketch or screenshot the spectrum and label the carrier and sideband frequencies and amplitudes. (5 points.)

4 Phasor Representation

As described in Section 1, the direct-conversion receiver (“Baba Yaga’s hut,” Figure 3) requires the local oscillator to be synchronized with the transmitter in frequency and in phase. The 10 MHz reference clock cable handles frequency synchronization. Phase must be calibrated manually, because the DDS generators don’t maintain phase lock.

An important practical consideration: the DDS generators shift phase in response to certain parameter changes but not others. Table 7 summarizes which actions are safe and which require recalibration.

Action	Phase Shift?	What To Do
Change AM Depth (e.g. 75% $\rightarrow$ 0%)	No	Safe to use.
Change carrier amplitude (e.g. $1 V_{pp} \rightarrow 0 V_{pp}$)	No	Safe to use.
Change carrier phase on XMTR	No	Safe to use.
Change AM Freq (e.g. 440 Hz $\rightarrow$ 1 kHz)	Yes	Recalibrate phasor afterwards.
Change modulation type (AM $\rightarrow$ DSB-AM)	Yes	Recalibrate phasor afterwards.
Toggle Mod button on/off	Yes	Recalibrate phasor afterwards.
Toggle Output button on/off	Yes	Recalibrate phasor afterwards.

Table 7: DDS phase behavior — what causes phase shifts and what doesn’t.

4.1 Calibrating the Phasor Display

1. **Establish the origin.** On the transmitter (Func Gen #1), set the carrier amplitude to $0 V_{pp}$ (Parameter $\mapsto$ Amplitude). The generator will clamp to its minimum of 2.0 mVpp — close enough to zero. The dot on Oscilloscope 2 now represents approximately true zero. Use the CH1 and CH2 **Position** knobs to center this dot on the screen. This sets the origin (0,0). Then set the carrier amplitude back to $1 V_{pp}$.
2. **Align the phasor.** With persistence on and AM depth at 75%, you should see a line on the display. Adjust the **Phase** parameter on the RCVR (Func Gen #2) to rotate this line until it starts at the origin and extends out along the positive X-axis. The far end of the line should be about **2.5 divisions** to the right of the origin at 50 mV/div.

If the line extends past the screen edge, increase VOLTS/DIV on both channels. If it is too short to see clearly, decrease VOLTS/DIV. If you see nothing, check that all four Output buttons are green and verify cabling.

If the phasor suddenly rotates or drifts at any point during the lab, repeat step 2 to recalibrate.

4.2 Phasor Experiments

With the phasor calibrated, explore how it responds to parameter changes:

Q4.2-1: Vary the modulation depth (Mod $\mapsto$ AM Depth) from 0% up to 120%. In general terms, how does the phasor change as you increase the depth? Sketch the phasor at a low and a high depth to illustrate. (You will make more detailed sketches in Section 6.) When done, return depth to 75%. (5 points.)

Q4.2-2: While observing the phasor on Oscilloscope 2, vary the phase on the transmitter (Parameter $\mapsto$ Phase) from 0° to 90° . (For best results, change 10 degrees at a time.) What do you observe in the phasor diagram? Describe with a labeled drawing. When done, return it to 0° . (5 points.)

Q4.2-3: Vary the carrier voltage (Parameter $\mapsto$ Amplitude) from $1 V_{pp}$ to $0.5 V_{pp}$. What do you observe in the phasor diagram? Describe with a labeled drawing. When done, return it to $1 V_{pp}$. (5 points.)

5 Demodulating to Audio

You should already hear the 440 Hz tone on the AM radio (if not, see Section 2.7). In this section you will explore what happens to the demodulated audio when you change the transmitter parameters.

Q5-1: On the transmitter (Func Gen #1), set the carrier amplitude (Parameter $\mapsto$ Amplitude) to $0 V_{pp}$, then back to $1 V_{pp}$. What do you hear in each case? (5 points.)

Q5-2: Now set the AM depth (Mod $\mapsto$ AM Depth) to 0%. You're now sending the radio an unmodulated AM carrier. What do you hear? Does it sound different from when you turned the signal off entirely? (5 points.)

Q5-3: Most AM radios include an automatic gain control (AGC) circuit, which automatically adjusts the receiver's gain to keep the audio output at a roughly constant volume despite variations in the received signal strength. Explain how the AGC may have affected your observations above. How might your observations have sounded different if this radio didn't have a working AGC? (5 points.)

Q5-4: Change the modulating frequency on the transmitter (Func Gen #1) using the dial, by about a Hz at a time, and write down your observations. (5 points.)

Q5-5: Set the modulating frequency back to 440 Hz. There's a tuning fork³ on the bench. The tuning fork is pitched at A440 — the same 440 Hz frequency we're using for our modulating signal. Strike the tuning fork and compare its pitch to the tone from the radio. Do you detect any pitch difference? (5 points.)

³Practice good tuning fork discipline — thwack it against something soft, like your hand or clavicle, not the edge of the table.

6 Varying Parameters: Modulation Depth

Before proceeding, return all transmitter settings to their defaults: AM Depth 75%, AM Freq 440 Hz, Amplitude 1 V_{pp} .

Now we will systematically vary the transmitter parameters and observe the effect across all four domains simultaneously.

Q6-1: On the **transmitter (Func Gen #1)**, vary the modulation depth as listed in Table 8. **Fill out Table 8, sketching the effect of the parameter changes on the time domain (oscilloscope), frequency domain (spectrum analyzer) and phasor views of the signal. Use color coding.** You've got a lot of screens to keep track of, so you may want to go through this once for each domain. Qualitative is fine. You may need to recalibrate the phasor display if it drifts. **(15 points.)**

Depth	Time Domain	Freq Domain	Phasor	Audio Description
0%				
50%				
100%				
120%				

Table 8: Varying modulation depth. *Default Parameters: Carrier frequency of 1260 kHz, carrier amplitude of 1 V_{pp} , modulating frequency of 440 Hz.*

7 Varying Modulation Frequency

Before proceeding, return the modulation depth to the default of 75%.

Q7-1: Fill out Table 9, sketching the effect of the parameter changes on the time domain (oscilloscope), frequency domain (spectrum analyzer) and phasor views of the signal. Use color coding. You should be qualitative, and reasonably quantitative, but not necessarily precise. **Note:** You will need to recalibrate the phasor display after changing the modulating frequency (see Table 7). You may also need to adjust the SEC/DIV on Oscilloscope 1 (see Section 2.5) and widen the spectrum analyzer span for the 5 kHz case (see Section 2.4). **(15 points.)**

Freq.	Time Domain	Freq Domain	Phasor	Audio Description
1 Hz				
440 Hz				
1 kHz				
5 kHz				

Table 9: Varying modulation frequency. *Default Parameters: Carrier frequency of 1260 kHz, carrier amplitude of 1 V_{pp}, modulation depth of 75%.*

8 Double Sideband Suppressed Carrier

Return the modulating frequency to 440 Hz. Change the modulation type on the transmitter from AM to DSB-AM⁴. **You will need to recalibrate the phasor display** after changing the modulation type (see Table 7).

Q8-1: Fill out Table 10. (10 points.)

Scheme	Time Domain	Freq Domain	Phasor	Audio Description
DSB-FC				
DSB-SC				

Table 10: AM (DSB-FC) versus AM (DSB-SC). *Default Parameters: Carrier frequency of 1260 kHz, carrier amplitude of 1 V_{pp} , modulation depth of 75%.*

⁴DSB-SC, really — double sideband, suppressed carrier

Q8-2: Which version of 440 Hz sounds closer to the correct pitch? (You may want to use the tuning fork, or an online tone generator—onlinetonegenerator.com, for instance—to gauge this.) (5 points.)

Q8-3: With a modulating frequency of 440 Hz, tune the radio a bit above and below the carrier frequency (or lack thereof) of the radio. What happens to the audio in each case? (5 points.)